



MD UMAR HUSSAIN | 2022462059

DEPARTMENT OF SCIENCE AND HUMANITIES

PHYSICS

(+91) 9362-503-780 | mdumarhussain12345@gmail.com



<https://www.linkedin.com/in/md-umar-hussain-05aa6a248/>

<https://www.instagram.com/umairhussain7777/>

OBJECTIVE

Aspiring researcher with a strong aptitude for innovative problem-solving and a passion for fundamental research. Seeking a challenging position to leverage my skills and cultivate a successful career

PERSONAL PROFILE

- **Father Name** : MD Ayub
- **Mother Name** : Shah Jahan Begum
- **Date of Birth** : 11 January 1999
- **Nationality** : Indian
- **Language Known** : English, Hindi, Nagamese
- **Address** : Public Ground DDSC Stadium Golaghat Road Dimapur Nagaland

SUMMARY OF QUALIFICATIONS

- Masters student in Physics at the National institute of Technology Nagaland, completed masters in May 2024.
- Personal interest, Knowledge and hands-on experience in Classical physics, Quantum physics, Electromagnetic waves, Electronics, Semiconductor, Solid State devices, Mathematics Physics, Numerical Methods, Experimental Techniques, Electromagnetism, Nano science, Nanotechnology, Nanomaterial's, and Nano Magnetism, Electronics, Polymer. With good knowledge of Semiconductor and Electromagnetism and project under Nanomagnetism and Nanoscience using VASP and WIN2K codes. The minor project was on copper ferrite (CuFe_2O_4) using DFT (Density Functional Theory). Major project was under MnGa having Phases of L1_0 -MnGa, D0_{22} - Mn_3Ga and L1_0 - $\text{Mn}_{1.66}\text{Ga}$.
- NET June 2024 score is 24, NET December 2023 score is 37, GATE February 2024 score is 7.
- Inquisitiveness in learning new things, creative approach to problem solving and ability to handle events and people.

RESEARCH INTEREST

1. Electromagnetism
2. Nano science
3. Electronics
4. Semiconductor
5. Solid State devices
6. Magnetism
7. Quantum physics
8. Classical physics



MD UMAR HUSSAIN | 2022462059

DEPARTMENT OF SCIENCE AND HUMANITIES

PHYSICS

(+91) 9362-503-780 | mdumarhussain12345@gmail.com



ACADEMIC QUALIFICATION

Year	Degree/Exam	Institution	Percentage
2022-2024	Master of Science	National Institute of Technology Nagaland	89%
2019-2022	Bachelor of Science	St. Joseph University	66%
2017-2019	Higher Secondary	Pranab Vidyapith Higher Secondary School	45%
2016-2017	Secondary	Zakiesato Memorial Higher Secondary School	86%

MASTERS PROJECT

- **Minor Project : FIRST PRINCIPLES STUDY ON THE STRUCTURAL, ELECTRONIC AND MAGNETIC PROPERTIES OF CuFe_2O_4**

ABSTRACT:

In this study, the electronic and magnetic properties of CuFe_2O_4 has been investigated by using first-principle calculations based on density functional theory (DFT). The electronic and magnetic properties such as density of states, band structure and magnetic moments has been analyzed to gain a comprehensive understanding of the material. The study revealed the insight understanding of the material's properties and its potential implications for the application in the field of materials science. This research gives the possibilities of understanding the CuFe_2O_4 and open up a new opportunities for explaining its distinctive structural and magnetic features.

Keywords: Density functional theory (DFT), permanent magnet, density of states(DOS), band structure, etc.

OBJECTIVES:

- To study the electronic structure and magnetic properties of CuFe_2O_4 using Density functional theory(DFT).
- Comparative study of different types of CuFe_2O_4 structures, their electronic and magnetic properties.

CONCLUSION:

In conclusion, by using First principle calculation we have done theoretical investigation on the electronic and magnetic properties of CuFe_2O_4 . In our study we found out that both the cubic and Monoclinic CuFe_2O_4 shows magnetic characteristic and metallic in nature. The calculated lattice parameters a, b, c and c/a ratio are in agreement with the earlier reported theoretical studies. The magnetic moment of both the alloys has been calculated and analyzed . Thus the overall studies shows that the materials can be applied in the field of materials science and technology.



MD UMAR HUSSAIN | 2022462059

DEPARTMENT OF SCIENCE AND HUMANITIES

PHYSICS

(+91) 9362-503-780 | mdumarhussain12345@gmail.com



➤ **Major Project: First-principle study on the electronic structure and magnetic properties of Different Phases of MnGa**

ABSTRACT: The L1₀-MnGa alloy is known for its high coercivity and remarkable magnetic anisotropy which is suitable for rare-earth free permanent magnet application. However, theoretically the material magnetic properties has not been explored much, which need to be investigated. As a result, in this work, using first-principle study based on density functional theory (DFT), the structural and magnetic properties of Different Phases of MnGa will be analyse in details. Furthermore, the influence of transition metals on the material properties will be investigated by substitution and doping into the MnGa alloy.

Keywords: Density functional theory (DFT), permanent magnet, density of states(DOS),MAE(Magnetic Anisotropic Energy), etc.

OBJECTIVES:

To study the electronic structure and magnetic properties of Different Phases of MnGa by using Density Functional Theory.

To study the magnetic properties of Different Phases of MnGa compounds by substitution of transition metals on Mn sites.

CONCLUSION:In conclusion, using first-principles calculations, we conducted a theoretical investigation on the electronic and magnetic properties of L1₀-MnGa. Our study revealed that L1₀-MnGa exhibits magnetic characteristics and metallic nature. The calculated lattice parameters (a, b, c) and c/a ratio align with previously reported theoretical studies. The magnetic moments of the alloys were calculated and analyzed, showing consistency with earlier reports. Specifically, the magnetic moment of the Mn atom in L1₀-MnGa is 2.549 μ_B , with a total magnetic moment of 4.826 μ_B . For L1₀-Mn_{1.66}Ga, the Mn atom's magnetic moment is 2.438 μ_B , with a total magnetic moment of 6.397 μ_B . In D022-Mn₃Ga, the Mn atom's magnetic moment is 2.310 μ_B , and the total magnetic moment is 0.648 μ_B . We examined the variation of magnetocrystalline anisotropy (MAE) and magnetization in Mn-Ga alloys, finding that MAE is significantly large in the studied alloys: MnGa, Mn₃Ga, and Mn_{1.66}Ga. The MAE values for L1₀-MnGa, L1₀-Mn_{1.66}Ga, and D022-Mn₃Ga are 2.21443 MJ/m³, 2.43836 MJ/m³, and 2.82284 MJ/m³, respectively. Overall, these findings demonstrate that these materials have promising applications in the field of materials science and technology.



MD UMAR HUSSAIN | 2022462059

DEPARTMENT OF SCIENCE AND HUMANITIES

PHYSICS

(+91) 9362-503-780 | mdumarhussain12345@gmail.com



BACHELORS PROJECT

➤ Major Project: PATIENT HEALTH MONITORING SYSTEM USING ARDUINO

Abstract: The presented research describes how to implement a system arduino-based system to monitor the patient health, For the last couple of decades ,simulating a system in practice team has been an important topic for the students community .As the students population is a learning on a daily basis ,we thought we were doing a projects which would entail some experiment and new software/technology to target .The project presented describe a system for monitoring, This system work by using arduino a microcontroller based on program.This is a arduino UNO R3 which is basically used for small devices.This used program intall in arduino UNO R3 with pluse sensor and temperature sensor fixed to it . The program in it is a PML languages which controls the microcontroller and sensor .this is used in any places and can take in place for using and monitoring the patient health .Therefore ,this system work on microcontroller which is based on program.

OBJECTIVES:

- a. It is used to check heart beat of a person
- b. It is used to check body temperature
- c. It is cost effective
- d. It is portable
- e. It is light-weight
- f. It is easy to used

Introduction:

With tons of new healthcare technology start-ups, IoT [1]is rapidly revolutionizing the healthcare [2]industry. In this project, we have designed the Patient Health Monitoring System The Arduino Sketch [4]running over the device implements the various functionalities of the project like reading sensor data, converting them into strings, passing them to the IoT platform, and displaying measured pulse rate and temperature[3] on character LCD. Arduino is an open-source hardware and software company, project,and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices. Its hardware 28products are licensed under a CC BY-SA license, while software is licensed under the GNU Lesser General Public License (LGPL) or the GNU General Public License (GPL),[1] permitting the manufacture of Arduino boards and software distribution by anyone. Arduino boards are available commercially from the official website or through authorized distributors.



MD UMAR HUSSAIN | 2022462059

DEPARTMENT OF SCIENCE AND HUMANITIES

PHYSICS

(+91) 9362-503-780 | mdumarhussain12345@gmail.com



SKILLS WITH EXPERIENCED

PROGRAMMING LANGUAGES	Year	Institute	Location
C	2019-2021	NITI (NATIONAL INSTITUTE OF TECHNOLOGY & INNVATION)	DIMAPUR NAGALAND
C++			
HTML			
CSS			
JavaScript			
Python			
SQL			
Graphics Designs	Duration	Institute	Location
Canvas	3 years	NITI (NATIONAL INSTITUTE OF TECHNOLOGY & INNVATION)	DIMAPUR NAGALAND
Adobe Photoshop			
Blender (3D modelling)			
Poster My Wall			
Computer Courses	Duration	Institute	Location
BASIC	3 years	NITI (NATIONAL INSTITUTE OF TECHNOLOGY & INNVATION)	DIMAPUR NAGALAND
DCA			
Hardware and Networking			
Web development			
Operating Systems	Duration	Institute	Location
Windows	3 years	NITI (NATIONAL INSTITUTE OF TECHNOLOGY & INNVATION)	DIMAPUR NAGALAND
Parrot os			
Kali			
Ubuntu			
Computational software	Duration	Institute	Location
VASP Code	2 years	NIT (NATIONAL INSTITUTE OF TECHNOLOGY NAGALAND)	CHUMOUKEDIMA NAGALAND
WIEN2K			
Originlab			



MD UMAR HUSSAIN | 2022462059

DEPARTMENT OF SCIENCE AND HUMANITIES

PHYSICS

(+91) 9362-503-780 | mdumarhussain12345@gmail.com



Programming Platform	Duration	Institute	Location
Sublime Text	3 years	NITI (NATIONAL INSTITUTE OF TECHNOLOGY & INNVATION)	DIMAPUR NAGALAND
VS code			
Notepad			
Python IDE			
Notepad++			
Turbo C++			
Dev C++			
Online compiler			
MYSQL Server			

COFERENCES ATTENDED

- Nominated candidate for the 73rd lindau Noble laureate meeting on Physics from 30 June -5 July 2024 (Germany)
- IEEE GCON 2023 Conference (IIT Guwahati)
Role : Participant
- 30th National Conference on Condensed Matter Physics, Condensed Matter Days - 2022(CMDAYS-22).(NIT Nagaland)
Role: Participant
- Electro spinning For Fabrication Of Nanofiber Memberane And Applications
Role: Participant. (NIT Nagaland)
- ALRIS 2023 (an occasion celebrating national science day)
Role: Participant (NIT Nagaland)
- National Webinar on battery Materials
Role : participant (St. Joseph university Nagaland)
- National conference on Smart materials and cartelization Techniques (NCSMCT-2019)
Role : Participant (St. Joseph university Nagaland)



MD UMAR HUSSAIN | 2022462059

DEPARTMENT OF SCIENCE AND HUMANITIES

PHYSICS

(+91) 9362-503-780 | mdumarhussain12345@gmail.com



POSITIONS OF RESPONSIBILITY

- ◆ **Organiser of Science Exhibition in National Science Day 2023**
- ◆ **Organiser of Innovation in National Science Day 2024**
- ◆ **Member of Science Society of NIT Nagaland 2023-2024**
- ◆ **Member of Robotics Club of NIT Nagaland 2023-2024**
- ◆ **Member of Coding Club of NIT Nagaland 2023-2024**
- ◆ **Faculty of National Academy of Computer Training and Skill Development 2024**

WORK AND EXPERIENCES

1. Technician	Mobile Network Jammer	part time	JEE-MAIN
2. Supervisor	Security	part-time	NEET UG
3. Invigilator	Exam duty	part-time	AGNIVEER
4. Technician	CCTV operator	part-time	JEE-MAIN
5. Computer Teacher	School students	1 years	Home
6. Coding Trainer	NACTSD	part-time	Dimapur



Scan for the References

I, hereby declare that all the details furnished above are true to the best of my knowledge and belief.